Vitamin D

Dentistry

VITAMIN D (CHOLECALCIFEROL)

There are reports of symptoms of rickets from historians as early as 2nd century AD. Francis Glisson wrote a classical account of infantile rickets in 1650. Experimental rickets induced by dietary deficiency was produced in rats by McCollum in 1919. Angus and coworkers isolated vitamin D in 1931 and named it as calciferol, which was later identified as Vitamin D3. The structural elucidation was done independently by Otto Diels and Kurt Alder. Both were awarded Nobel prize in 1950.

Formation of Vitamin D

Vitamin D is derived either from 7-dehydro-cholesterol or ergosterol by the action of ultraviolet radiations. **7-dehydrocholesterol**, an intermediate of a minor pathway of cholesterol synthesis, is available in the Malpighian layer of epidermis. In the skin, ultraviolet light (290-315 nm) breaks the bond between position 9 and 10 of the steroid ring. So, the ring B is opened, to form the provitamin, **secosterol** (Fig. 33.8). The cis double bond between 5th and 6th carbon atoms, is then isomerized to a trans double bond (rotation on the 6th carbon atom) to give rise to vitamin D_3 or **cholecalciferol** (Fig. 33.8). So, vitamin D is called the **"sun-shine vitamin".**

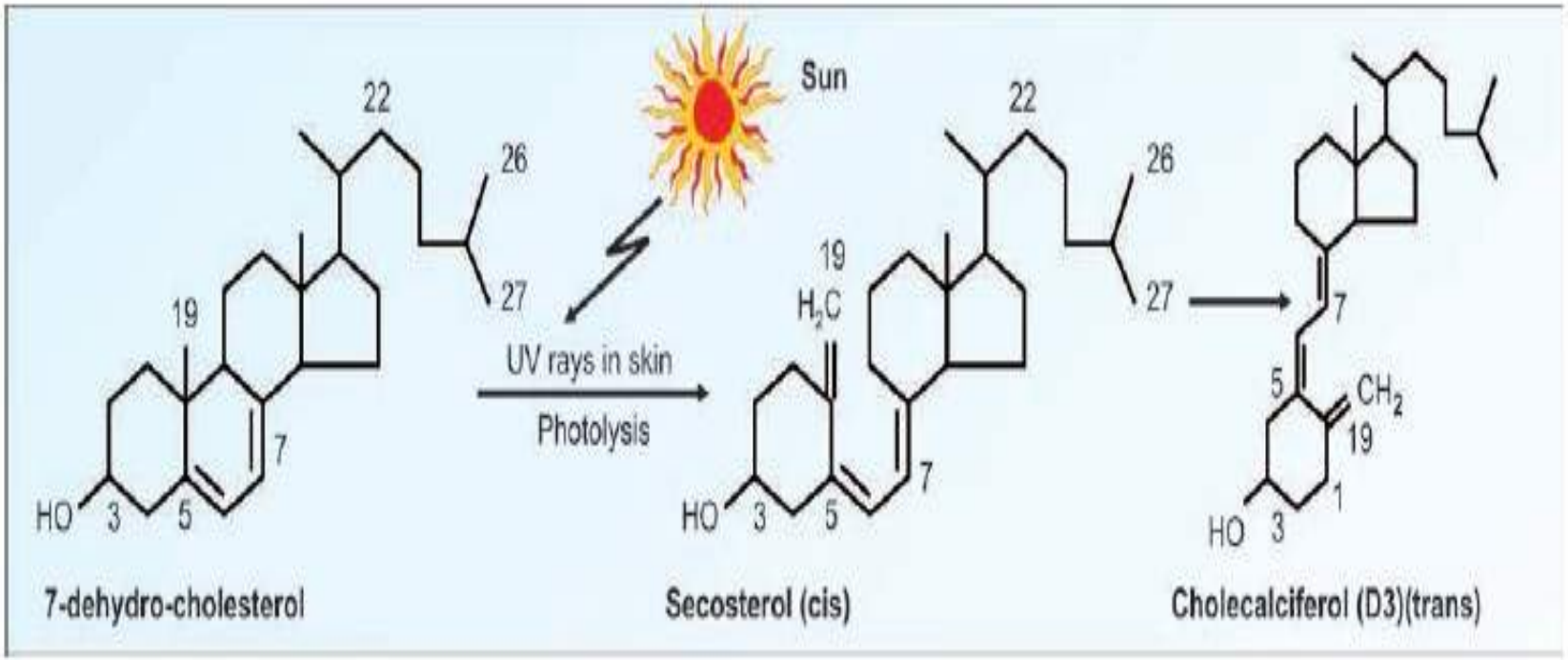


Fig. 33.8. Synthesis of cholecalciferol or vitamin D3

Skin is the largest organ in the body. It makes about 16% of body weight. It is kept nourished by a quarter of the body's blood supply. It is the temperature regulator of the body. The production of vitamin D in the skin is directly proportional to the exposure to sunlight and inversely proportional to the pigmentation of skin. An increase in solar zenith angle during November to March shifts the wavelength of UV rays to longer wavelengths which will not produce the vitamin; hence vitamin deficiency is seen in winter. Excessive exposure to sunlight does not result in vitamin D toxicity since excess previtamin D_3 and D_3 are destroyed by sunlight itself.

Commercially the vitamin is derived from the fungus, ergot. The ergosterol when treated with ultraviolet light, **ergocalciferol** or vitamin D_2 is produced. Ergocalciferol differs in having an unsaturation in the side chain and an extra methyl group (C28).

Activation of Vitamin D

i. Vitamin D is a **prohormone**. The cholecalciferol is first transported to **liver**, where hydroxylation at **25th position** occurs, to form 25-hydroxy cholecalciferol (25-HCC). The hepatic 25-hydroxylase is a microsomal monooxygenase. It requires cytochrome P-450 and NADPH (Fig. 33.9). 25-HCC is the major storage form.

ii. In plasma, 25-HCC is bound to "vitamin D binding protein" (VDBP), an alpha-2 globulin.

iii. In the **kidney**, it is further hydroxylated at the **1st position**. The 1-alpha hydroxylase is located in mitochondria of proximal convoluted tubules. It requires cytochrome P-450, NADPH and ferrodoxin (an iron-sulfur protein). Thus 1,25-dihydroxy cholecalciferol (**DHCC**) is generated. Since it contains three hydroxyl groups at 1, 3 and 25 positions, it is also called **Calcitriol** (Fig. 33.9). The calcitriol thus formed is the **active form** of vitamin; it is a **hormone** (See Box 33.1). 24,25-dihydroxy cholecalciferol may be formed by hydroxylation of 25-HCC at the 24th position.

Box 33.1. Calcitriol and Calcitonin are Different

Calcitriol is the physiological active form of vitamin D. It increases the blood calcium level.

Calcitonin is the peptide hormone released from thyroid gland. It decreases the blood calcium.

Biochemical Effects of Vitamin D

The sites of action are:

a. intestinal villi cells
b. bone osteoblasts
c. kidney distal tubular cells.

A. Vitamin D and Absorption of Calcium

Calcitriol promotes the absorption of calcium and phosphorus from the intestine. In the brush-border surface, calcium is absorbed passively. From the intestinal cell to blood, absorption of calcium needs energy. It is either by the sodium-calcium exchange mechanism or by pumping out the calcium-calbindin complex. **Calcitriol** acts like a steroid hormone. It enters the target cell and binds to a cytoplasmic receptor. The hormone-receptor complex interacts with DNA and causes **derepression** and consequent transcription of specific genes that code for **Calbindin** (Fig. 33.10). Due to the increased availability of calcium binding protein, the absorption of calcium is increased.

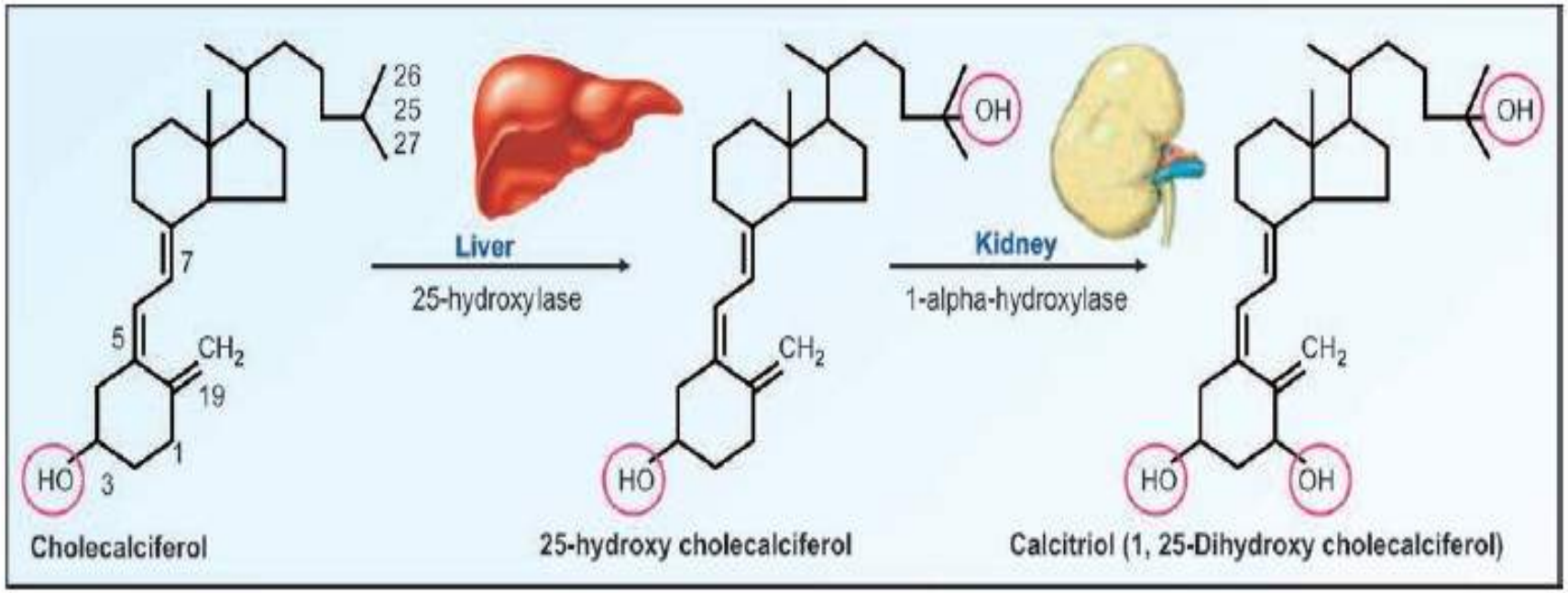


Fig. 33.9. Generation of calcitriol

B. Effect of Vitamin D in Bone

Mineralization of the bone is increased by increasing the activity of osteoblasts (Chapter 35). Calcitriol coordinates the remodelling action of osteoclasts and osteoblasts. It produces the differentiation of osteoclast precursors from multinucleated cells of osteoblast lineage. Calcitriol stimulates osteoblasts which secrete alkaline phosphatase. Due to this enzyme, the local concentration of phosphate is increased. The ionic product of calcium and phosphorus increases, leading to mineralization.

C. Effect of Vitamin D in Renal Tubules

Calcitriol increases the reabsorption of calcium and phosphorus by renal tubules, therefore both minerals are conserved (PTH conserves only calcium) (Chapter 35).

D. Regulation of Calcitriol

The hormonal level of calcitriol is maintained by the feed-back control. The rate of production is modulated by serum levels of calcium, phosphorus, PTH and calcitriol itself. The major site of control is on the 1-alpha-hydroxylase of kidney. Hypercalcemia decreases calcitriol. Low dietary calcium and hypocalcemia increase the rate of production of 1,25-DHCC. The stimulatory effect of hypocalcemia on 1-alpha-hydroxylase is through PTH. Hypercalcemia decreases calcitriol. The half-life of 1, 25-DHCC is 6-8 hours and is

mainly excreted through the bile. The inactivation is by oxidation of the side chain. Hydroxylation at the 24th position by the kidney also lowers the activity. Vitamin D deficient state (low calcium and phosphate) will enhance the action of 1-alpha hydroxylase. When vitamin D status is adequate, or calcium level is high, 24-hydroxylation is favored, forming 24,25-DHCC, which is less active. 1,25-DHCC level directly regulates PTH secretion.

Deficiency of Vitamin D

The deficiency diseases are **rickets** in children and **osteomalacia** in adults. Hence vitamin D is known as antirachitic vitamin.

A. Causes for Vitamin D Deficiency

i. Deficiency of vitamin D can occur in people who are not exposed to sunlight properly, e.g. inhabitants of northern latitudes, in winter months, in people who are bedridden for long periods, or those who cover the whole body *(purdah)*.

ii. Nutritional deficiency of calcium or phosphate may also produce similar clinical picture.

iii. Malabsorption of vitamin (obstructive jaundice and steatorrhea). High phytate content in diet may also reduce the absorption of vitamin.

iv. Abnormality of vitamin D activation. Liver and renal diseases may retard the hydroxylation reactions.

v. Deficient renal absorption of phosphates.

B. Clinical Features of Rickets

i. Rickets is seen in children. There is insufficient mineralization of bone. Bones become soft and pliable. The bone growth is markedly affected.

ii. Plasma calcium and phosphorus are low normal with alkaline phosphatase (bone iso-enzyme) being markedly elevated.

iii. The classical features of rickets are **bone deformities**. Weight bearing bones are bent (Fig. 33.11). Continued action of muscles also cause bone malformations.

iv. The clinical manifestations include bow legs, knock-knee, rickety rosary, bossing of frontal bones, and pigeon chest.

v. An enlargement of the epiphysis at the lower end of ribs and costochondral junction leads to beading of ribs or **rickety rosary**.

vi. **Harrison's sulcus** is a transverse depression passing outwards from the costal cartilage to axilla. This is due to the indentation of lower ribs at the site of the attachment of diaphragm.



Fig. 33.11. Bone deformity in rickets

D. Different Types of Rickets

i. The classical vitamin D **deficiency** rickets can be cured by giving vitamin D in the diet.

ii. The **hypophosphatemic** rickets mainly result from defective renal tubular reabsorption of phosphate. Supplementation of vitamin D along with phosphate is found to be useful.

iii. **Vitamin D resistant** rickets is found to be associated with *Fanconi syndrome*, where the renal tubular reabsorption of bicarbonate, phosphate, glucose and amino acids are also deficient.

iv. **Renal rickets**: In kidney diseases, even if vitamin D is available, calcitriol is not synthesized. These cases will respond to administration of calcitriol.

v. **End organ refractoriness** to 1,25-DHCC will also lead to rickets. Either a decrease in the number of cytosolic receptor or a structurally abnormal receptor is noticed. The bone disease has been found to respond to megadoses of calcitriol (35 mg/day).

Other Actions of Vitamin D

Recent research has proved that most tissues possess vitamin D receptor and several tissues have the enzymes to generate calcitriol. 1,25-DHCC has been found to have a modulatory effect on immunohematopoietic system. Therapeutic doses given to children with rickets have been found to correct the anemia, hypocellularity of the bone marrow and increased susceptibility to infection. It has also been found to reduce the risk of cancer and coronary vascular disease. Beneficial effects have been observed in patients with AIDS.

Requirement of Vitamin D

i. Children = 10 microgram (400 IU)/day
ii. Adults = 5 microgram (200 IU)/day
iii. Pregnancy, lactation = 10 microgram/day
iv. Above the age of 60 = 600 IU per day.

Sources of Vitamin D

Exposure to **sunlight** produces cholecalciferol. Moreover fish liver oil, fish and egg yolk are good sources of the vitamin. Milk contains moderate quantity of the vitamin.

Hypervitaminosis D

Doses above 1500 units per day for very long periods may cause toxicity. Symptoms include weakness, polyuria, intense thirst, difficulty in speaking, hypertension and weight loss. Hypercalcemia leads to calcification of soft tissues, (metastatic **calcification**, otherwise called **calcinosis**), especially in vascular and renal tissues. Although vitamin D is toxic in higher doses, excessive exposure to sunlight does not result in vitamin D toxicity, because excess D_3 is destroyed by sunlight itself.



Oral Health

- Diet and nutrition play a key role in
  - —Tooth development
  - —Gingival and oral tissue integrity
  - —Bone strength
  - —Prevention and management of diseases of the oral cavity

TABLE 28-1 Effects of Nutrient Deficiencies on Tooth Development

NUTRIENT	EFFECT ON TISSUE	EFFECT ON CARIES	HUMAN DATA
Protein/calorie malnutrition	Delayed tooth eruption Decreased tooth size Decreased enamel solubility Salivary gland dysfunction	Yes	Yes
Vitamin A	Decreased epithelial tissue development Tooth morphogenesis dysfunction Decreased odontoblast differentiation Increased enamel hypoplasia	Yes	Yes
Vitamin D/calcium/phosphorus	Lowered plasma calcium levels Hypomineralization (hypoplastic defects) Compromised tooth integrity (decreased mineral concentration) Delayed eruption patterns	Yes	Yes
Ascorbic acid	Dental pulpal alterations Odontoblastic degeneration Aberrant dentin	No	No
Fluoride	Increased stability of enamel crystal (enamel formation) Inhibition of demineralization Stimulation of remineralization Mottled enamel (excess) Inhibition of bacterial growth	Yes	Yes
Iodine	Delayed tooth eruption Altered growth patterns Malocclusions?	No	Yes
Iron	Slow growth Salivary gland dysfunction	Yes	No

Modified from DePaola D, Faine MP, Vogel RI: Nutrition in relation to dental medicine. In Shils ME, Olson JA, Shike M, editors: *Modern nutrition in health and disease*, vol 2, ed 8, Philadelphia, 1994, Lea & Febiger.

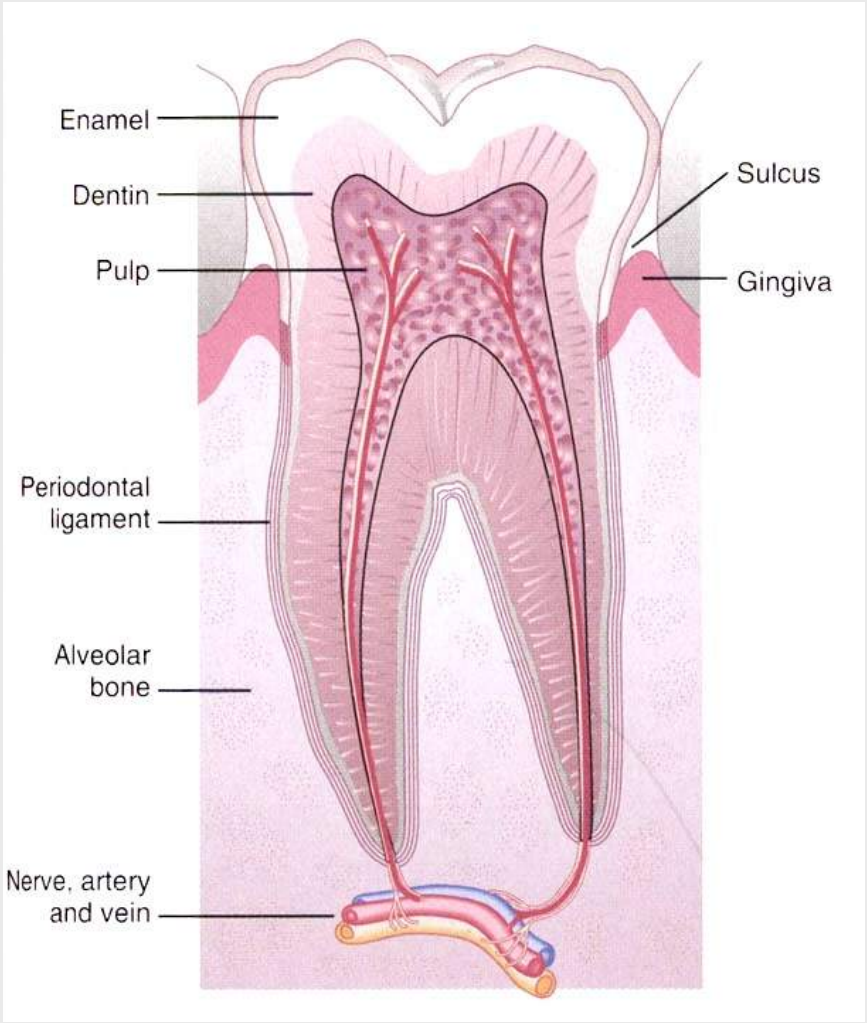


Dental and Oral Health

- Teeth are made from protein matrix that is mineralized with collagen (requiring vitamin C), calcium, and phosphorus (requiring vitamins D and A)

Anatomy of a Tooth







Vitamin D’s role in oral health

Recent research, especially by Dr Robert Schroth in Canada,has linked vitamin D deficiency in pregnancy to enamel hypoplasia (figure 1) of the offspring, a risk factor for tooth decay.

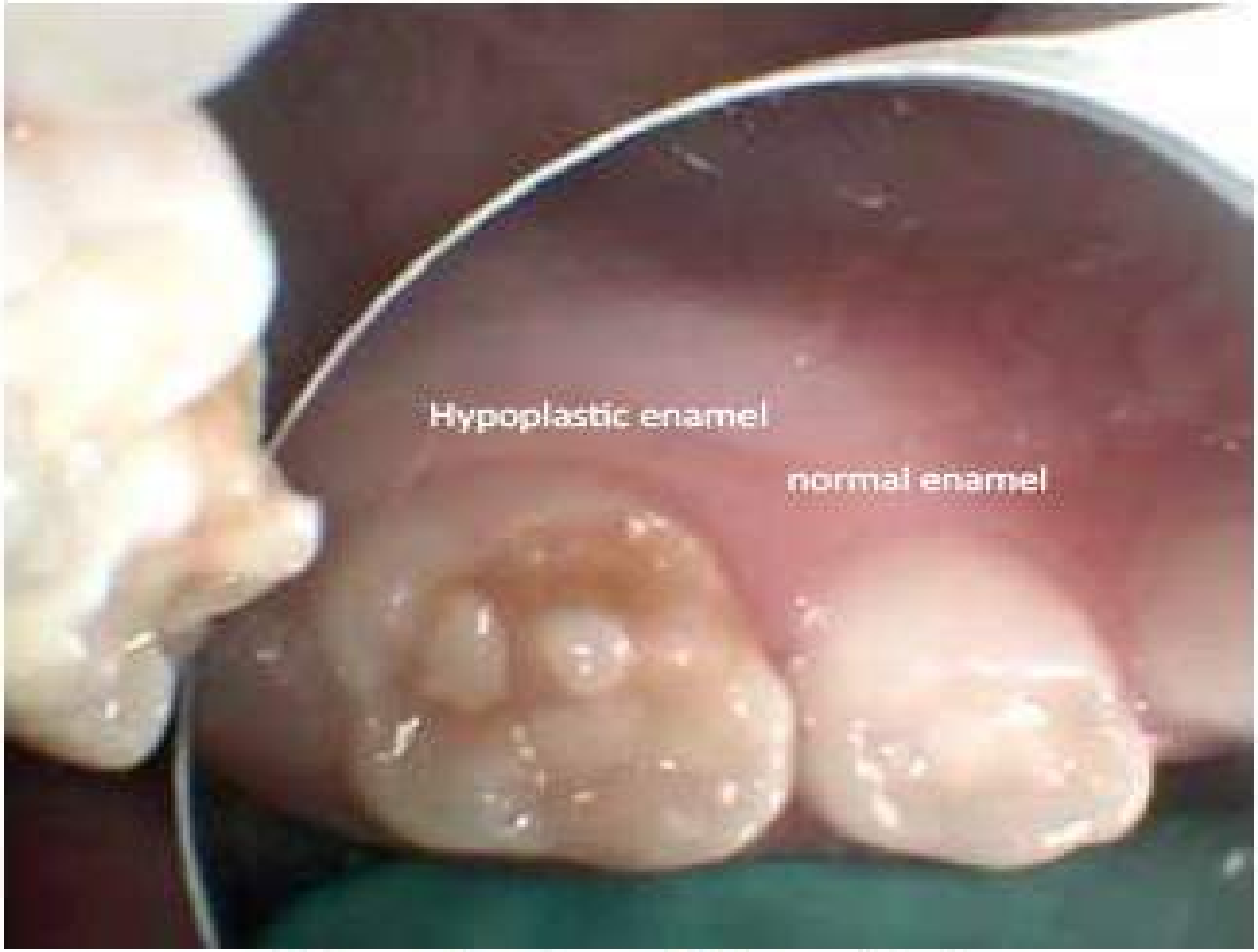


Figure 1. Enamel hypoplasia



Vitamin D plays a key role in craniofacial development and the maintenance of good oral health (Berdal et al. 2005). There is a growing body of evidence regarding the association between vitamin D and dental health (Dietrich et al. 2004; Grant 2008; Schroth et al. 2014). Inadequate concentrations of 25-hydroxyvitamin D (25(OH)D) have been linked with periodontal disease, tooth loss, and oral bone loss (Dietrich et al. 2004), and vitamin D supplementation has been shown to improve clinical periodontal outcomes (Krall et al. 2001; Hildebolt 2005). Vitamin D also influences enamel and dentin formation (Berdal et al. 2005). Episodes of malnutrition and vitamin D deficiency during periods of primary and permanent tooth formation can result in enamel hypoplasia and dental caries (Purvis et al. 1973; Alvarez et al. 1993).



- When one considers that the teeth, including the permanent ones, start developing in utero, the need to ensure our pregnant and lactating moms have optimal vitamin D status should be the aim of all health professionals in the perinatal arena. Two recent studies 4,5 from the US have highlighted vitamin D deficiency: one study showed 76% of mothers giving birth and 81% of their newborns were deficient (< 20ng/mL).



Delayed Tooth Eruption

Delayed tooth eruption in a child is a worrisome experience for parents and it necessitates early investigations and interventions. There is rare literature or reports on delayed eruption of teeth due to vitamin D deficiency. This particular case is very interesting because delayed tooth eruption was the sole feature in this patient and without any other clinical symptoms of Vitamin D deficiency.

Reference

Delayed Teeth Eruption A Mirror To Systemic Deficiency Of Vitamin D

Teeth structure

If some rich source of vitamin D be added, such as cod-liver oil or egg-yolk, the structure of the teeth will be greatly improved, while the addition of oils such as olive or arachis oil leaves the teeth as badly formed as when the basal diet only is given (Fig. 11). Not only are the defects in the dentine and enamel obvious on microscopic examination, but external examination reveals the surface enamel also to be badly formed (Fig. 12). If, when the vitamin D intake is deficient, the cereal part of the diet is increased, or if wheat germ replaces part of the white flour (Fig. 13 (*a*)), or, again, if oatmeal is substituted for white flour (Fig. 13 (*e*)), then the teeth tend to be worse in structure, but if, under these conditions, the calcium intake is increased, then calcification is improved (Fig. 14). Calcium appears both to antagonise the anticalcifying effect of cereals and to aid the action of any vitamin D present in the diet, and this action becomes of great importance if butter is the fat of the diet. Butter in itself has a comparatively



small calcifying influence in the presence of oatmeal, but a corresponding quantity of fat given as milk instead of butter, or an additional amount of calcium with the butter, may result in the development of perfectly calcified teeth (Fig. 15 (*a*), (*b*) and (*c*)).

It is obvious, therefore, that, although vitamin D holds the key-position, dental structure and incidentally bone formation cannot in actual practice be regarded as controlled by only one factor, but that these developmental processes are complicated mechanisms controlled by a series of interactions, some favouring and some antagonising perfect architecture. In addition, it may be stated that, since the earlier stages of calcification are more unstable than the later ones, it is most important to get the beneficial influences to work as early as possible; and this can be accomplished only by suitably feeding the mother during pregnancy and the offspring in its early months of life.

Reference

NUTRITION AND DISEASE

THE INTERACTION OF CLINICAL AND EXPERIMENTAL WORK



Vitamin D and Dental Caries in Children

Data from a cross-sectional, nationally representative sample of Canadian children suggest that there is an association between dental caries and lower vitamin D. Children with 25(OH)D concentrations ≥75 nmol/L had a 39% lower odds of having experienced caries, while children with levels ≥50 nmol/L had a 47% lower odds for caries. Improving children's vitamin D status may be an additional preventive consideration to lower the risk for caries. Recommending regular vitamin D supplementation during childhood may decrease the overall burden of tooth decay in children.

Reference: *Vitamin D and Dental Caries in Children*



POTENTIAL TARGET-GENES FOR 1,25 IN DEVELOPING TOOTH

Involvement in tooth germ	Expression in teeth		Control by 1,25 in other tissues	
Cell growth and differentiation				
	EGFr	Davideau *et al.*, in press	breast carcinoma	Desprez *et al.*, 1991
Growth factors	*NGF*	Mitsiadis and Byers, this issue	Fibroblast L-292	Wion *et al.*, 1991
and receptors	*TGFß*	D'Souza *et al.*, 1990	Bone	Finkelman *et al.*, 1991
	BMP	Bègue-Kirn *et al.*, 1992		
Proto-oncogenes	*c-fos*	Caubet and Bernaudin, 1988	Osteoblastic cells	St. Arnaud *et al.*, 1994
	c-myc	Hirning *et al.*, 1992	Keratinocytes	Hanafin *et al.*, 1994
Factors controlling transcription	*Msx-2*	McKenzie *et al.*, 1992	Bone	Hodgkinson *et al.*, 1993
	VDR	Berdal *et al.*, 1993	Bone	Arbour *et al.*, 1993
Extracellular matrix	*Fibronectin*	review: Lesot, 1986 Thesleff *et al.*, 1991	Fibroblasts	Franceschi *et al.*, 1987
	α1(I) collagen	Andujar *et al.*, 1991		Lichtler *et al.*, 1989
Hard tissue formation				
	α1(I) collagen	Pavlin *et al.*, 1992	Bone	Lichtler *et al.*, 1989
Matrix secretion	*Osteocalcin*	Bronckers *et al.*, 1994	Bone	Ozono *et al.*, 1991
	Osteopontin			Oldberg *et al.*, 1990
Ionic handling during mineralization	*Calbindins*	Berdal *et al.*, 1993	Intestine, kidney	for review: Lowe *et al.*, 1992
	Calcium pump	Borke *et al.*, 1993	Intestine	Wasserman *et al.*, 1994
	Alkaline phosphatase	Engström *et al.*, 1977	Bone	Kyeyune-Nyombi *et al.*, 1989

Reference

Ameloblasts and odontoblasts, target-cells for 1,25-dihydroxyvitamin D3: a review



Know your status

The only way to detect sufficiency is by a simple blood test. It's called a serum assay of 25(OH)D. Results below 20 ng/mL are 'deficient'. (At this level, babies and infants can present with rickets; in adults it's osteomalacia, a generalised painful bone disease with muscle weakness.) 35-40ng/mL is considered sufficient but many experts consider 50-100ng/mL the optimal concentration for protecting against cancer and heart disease.[2]



Guideline table of suggested supplements:		
Age range	Adequate intake (IU/day)	Safe Upper Limit (IU/day)
0 - 1 year old	400 - 1000	2000
1 - 12 years old	1000 - 2000	5000
13+ years old	1500 - 2000	10 000
Obese individuals	2 - 3 times above values	
Pregnant women	1400 - 2000	10 000
Nursing mothers *	2000 - 4000	10 000

* Nursing Mothers who want to ensure that their baby is getting enough vitamin D from their breast milk should take 4,000 – 6,000 IU/day.[6]

Reference: In the dark about vitamin D?